

A comparative analysis of Von-Heijne, SVM and Perceptron Models for the detection of Signal Peptides.

Abstract

Motivation: This is the motivation

Results: This are the results.

Introduction

For a protein to enter the secretory pathway, in both eukaryotic and prokaryotic cells, it must be endowed with a specific target signal. Often, this signal takes the shape of a short sequences located at the N-terminus of proteins ref. The signal peptide (SP) has a distinct three-domain structure, depicted in Figure 1, with a positively charged N-terminal region (n-region), a central hydrophobic region (H-region) and a more polar C-terminal region (C-region) containing the cleavage site ref. Given their importance in many aspects of cell biology, the accurate detection of SPs is a crucial task in bioinformatics, which has been tackled before with a variety of approaches, including machine learning (SVMs ref, ref and Bayesian networks ref, Hidden Markov Models ref), and, more recently, deep learning ref.

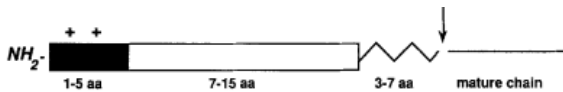


Figure 1: Signal peptide structure

With this work, we propose a comparative analysis of two different approaches for the detection of SPs, namely the Von-Heijne model a Support Vector Machine (SVM) based method.

Materials and Methods

Data Collection

Reviewed non-fragment protein sequences from the eukaryota taxa were queried from UniprotKB, and collected into 2 distinct sets, positive and negative. For Positive sequences, we selected sequences having experimental evidence for the presence of a signal peptide (ft_signal_exp:*) For the negative set, proteins not annotated with a signal peptide and endowed with experimental evidence for subcellular localization in non-secretory compartments, namely nucleus (cc_scl_term_exp:SL-0191), peroxisome (cc_scl_term_exp:SL-0204), cell membrane /

plasma membrane (cc_scl_term_exp:SL-0039), cytosol (cc_scl_term_exp:SL-0091), plastid (cc_scl_term_exp:SL-0209) or mitochondrion (cc_scl_term_exp:SL-0173). For sequences in the negative set, we gathered information about the presence of a transmembrane (TM) helix, whose hydrophobicity profile mimicks the one of a signal peptide.

Further selection on the positive entries was carried out in order to select proteins which had informaton about the cleavage site and a Signal Peptide with a length of at least 14bp.

Clustering

Sequences were clustered using mmseqs2 ref, using parameters --min-seq-id 0.3, cluster-mode 1 -cov-mode 0 -c 0.4, in order to cluster sequences with a similarity of at least 30% and an alignment spanning at least 40% of the total sequence length. The number of sequences in the datasets is reported in Table 1.

Dataset	Before Clustering	After Clustering
Positive	2942	1093
Negative (total)	20806	9028
Negative (w/o TM)	18301	8113
Negative (with TM)	2505	915

Table 1: Dataset sizes before and after clustering

Von-Heijne Model

The full dataset was split into a training and a test set with a ratio of 0.8. A Position-Specific Weight Matrix (PSWM) was then generated using 16bp windows (13bp upstream to 2bp downstream) surrounding the cleavage site. Amino acid frequencies were normalized using background amino acid frequencies from SwissProt. After initializing the matrix with pseudocounts of 1, the weights are computed as follows:

$$w(a, i) = \log_2 \left(\frac{f(a, i)}{f(a)} \right)$$

Where $f(a, i)$ is the frequency of amino acid a at position i in the training set, and $f(a)$ is the background frequency of amino acid a in SwissProt. The score for a given sequence is then computed as:

$$S = \max_{j=1}^{90-16+1} \sum_{i=1}^{16} w(a_{j+i-1}, i)$$

scores are computed over the first 90bp of the given sequence, using a sliding window 16bp long. Then, a sequence was labeled as positively endowed with a SP if the Score S is greater than a threshold t .

$$\text{label} = \begin{cases} \text{positive} & \text{if } S > t \\ \text{negative} & \text{if } S \leq t \end{cases}$$

The optimal threshold for the dataset was computed using 5-fold cross validation.

Feature Extraction

To build a model that can deal with sequences, they must be encoded into numerical representation. There are many possibility of extracting meaningful numerical features from a sequence, for the SVM and MLP model, we employed 2 kinds of features:

- Aminoacid composition up to a cut-off k (n_term_composition) and from k to an hardcoded limit N (set to 90) (c_term_composition)
- Protein Scales computed on the first N aminoacids with a window size of 5 (Figure 2). An Hydrophobicity scale (cit), alpha-helix tendency scale (cit), transmembrane tendency (cit), bulkiness (cit), and polarity (cit) were used.

For this experiment, k was chosen to be the average position of the cut site, roughly 22 aminoacids from the N-Terminus.

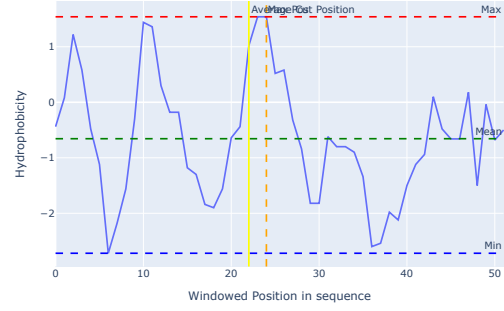


Figure 2: Example of feature extraction for a given sequence. Protein scales are computed on the first N aminoacids with a window size of 5

Support Vector Machine

We used a grid search to find optimal hyperparameters, testing both RBF and linear kernels. For the RBF kernel, we evaluated C values of 0.1, 1, 2, 4, and 8, combined with gamma values of “scale”, 0.001, 0.01, 0.1, 1, and 2. For the linear kernel, we tested C values of 0.1, 1, 2, 4, and 8. Each model was tested on a and evaluated on an 80% training split. The model yielding the highest f1 score was found to have a rbf kernel, with $C = 2$ and $\gamma = 0.01$, with a mean f1 score across the 5 folds ran on the training set of 0.87.

Evaluation and comparison

To evaluate and compare the performance of the models, we computed several metrics on the test set. The metrics were calculated based on the following formulas:

Accuracy: The proportion of correctly classified instances among all instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: The proportion of true positive predictions among all positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity): The proportion of true positive instances that were correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1 Score: The harmonic mean of precision and recall.

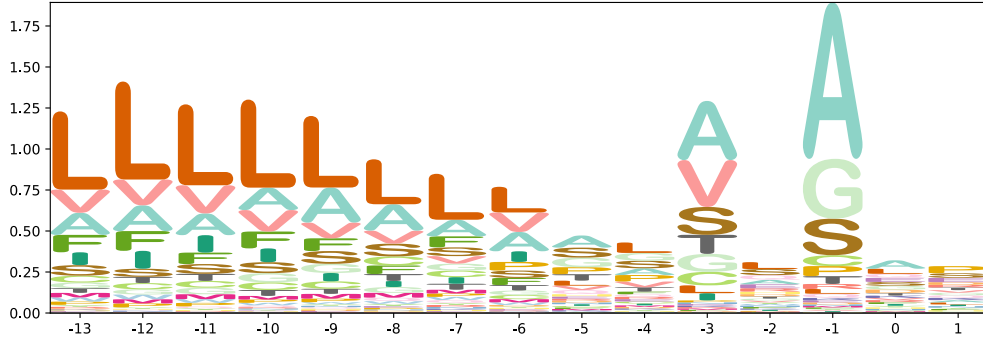


Figure 4: Motif logo for the context (-13bp to 2bp) surrounding the cleavage site

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$$F_1 = 2 \cdot \frac{(\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}$$

Matthews Correlation Coefficient (MCC): A balanced measure that takes into account all four confusion matrix categories.

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Where TP (True Positives), TN (True Negatives), FP (False Positives), and FN (False Negatives) are the values from the confusion matrix.

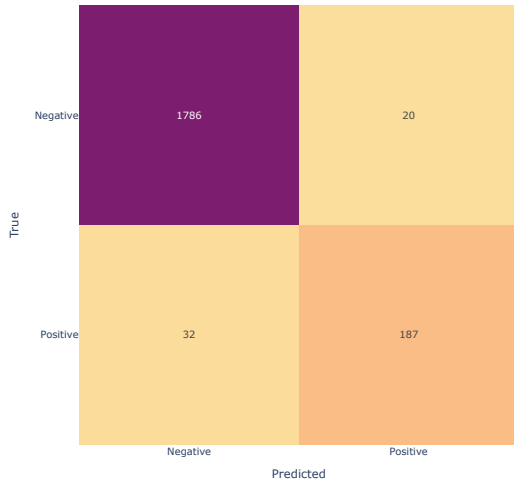


Figure 3: Confusion matrix for the SVM model

Results

SP Motifs

From all SP-Endowed sequences, a motif logo was generated using the context around the cleavage site (13 bp upstream to 2bp downstream) Figure 4.

For the Von-Hejne model the computed optimal threshold was 9.25, confusion matrix for the test set is reported in Figure 5.

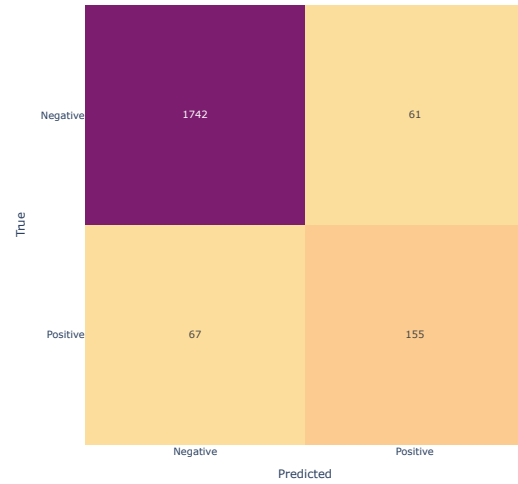


Figure 5: Confusion matrix for the Von-Hejne model